

Student Name: Mihir Ragoonauth
Student Number: M00955111
Submission Advisor: Mehnaz Hamilton

Project Gantt Chart

Task Name	January		February				March				April	
	Week 13 19/01/26	Week 14 26/01/26	Week 15 02/02/26	Week 16 09/02/26	Week 17 16/02/26	Week 18 23/02/26	Week 19 02/03/26	Week 20 09/03/26	Week 21 16/03/26	Week 22 23/03/26	Week 23 30/03/26	Week 24 06/04/26
Research & Planning												
Project scope definition and requirements analysis												
Literature review (thrust reversers, digital twins, AI fault detection)												
CAD model sourcing and purchase (Cults3D Trent 900)												
System architecture and methodology definition												
Digital Twin Development												
Siemens NX MCD environment setup and CAD import												
Rigid bodies, collision bodies and kinematic joints												
Position controls and motion simulation (52mm at 20.8mm/s)												
Virtual sensor configuration (8 channels, 100Hz)												
16-parameter Signal Adapter configuration												
GLB export for Three.js 3D viewer												
Control System Integration												
TIA Portal V19 project setup and SCL programming												
S7-PLCSIM Advanced 6.0 connection via .NET API (pythonnet)												
State machine development (States 10-80, Timer_2 15s)												
State 40 continuous parameter copy for mid-cycle fault injection												
6 AI PLC tags added for classifier feedback												
Data Generation & AI Development												
Automated data logging via .NET API (TD-007)												
250-scenario batch recording (5 fault classes)												
Feature extraction (45 features, Random Forest 91.8%)												
Algorithm comparison (RF, NN, Gradient Boosting, XGBoost)												
1,400-scenario expansion (7 classes incl. Oscillating, Stall)												
65-feature extraction with per-engine velocity masking fix												
XGBoost 7-class classifier (99.8% accuracy, 24.7ms latency)												
10-run robustness evaluation and model validation												
Operator Interface Development												
CustomTkinter GUI V1-V15 (EICAS, System Overview, Fault Injection)												
PyQt5 rebuild V1-V7 (native high-DPI scaling, TD-014)												
Three.js 3D engine schematic in QWebEngineView (TD-016)												
Landing simulation tab (A380 physics model)												
Audio alerts (stall 780Hz, GPWS horn, engine rumble 80Hz)												
V9 final operator interface with 5 tabs												
Physical Station (Festo EduTrainer)												
FluidSim pneumatic circuit design and simulation												
Station setup, wiring and PLC connection (S7-1200)												
Tag table corrections (double solenoid discovery)												
SCL function blocks and WinCC HMI on KTP700												
V1 station: 4 fault scenarios tested												
snap7 physical PLC integration attempt (TD-018)												
V2 station: rotary actuators, 3-position sensing, 7 scenarios (TD-019)												
Documentation & Submission												
Technical blog updates (ongoing)												
Technical decisions documentation (TD-001 to TD-019)												
Risk assessment (original and physical station update)												
Technical report (Chapters 1-6, 7,461 words)												
Presentation preparation (addressing mock feedback)												
Demonstration video recording												
Final submission												